



Minggu ke-6

Praktikum Decision Tree

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Klasifikasi dengan Decision Tree

```
from sklearn.tree import DecisionTreeClassifier  
  
...  
  
dtc=DecisionTreeClassifier()  
dtc.fit(train_data, train_label)  
class_result=dtc.predict(test_data)  
  
...  
  
acc=dtc.score(train_data, train_label)
```

Menampilkan Gambar Decision Tree

```
from sklearn import tree
import graphviz

...

dot_data = tree.export_graphviz(dtc, out_file=None,
                                feature_names=train_data.columns.values)

graph = graphviz.Source(dot_data, format="png")
graph.render(view=True)
```

Contoh Klasifikasi pada Ruspini Dataset

```
import pandas as pd
from sklearn import tree
from sklearn.tree import DecisionTreeClassifier
import graphviz
```

```
dataset = pd.read_csv('ruspini.csv')
```

```
train_data=dataset[['X', 'Y']]
train_label=dataset[['CLASS']]
```

```
dtc=DecisionTreeClassifier()
dtc.fit(train_data, train_label)
test_data=[[130, 45]]
class_result=dtc.predict(test_data)
print('Class = ', class_result)
```

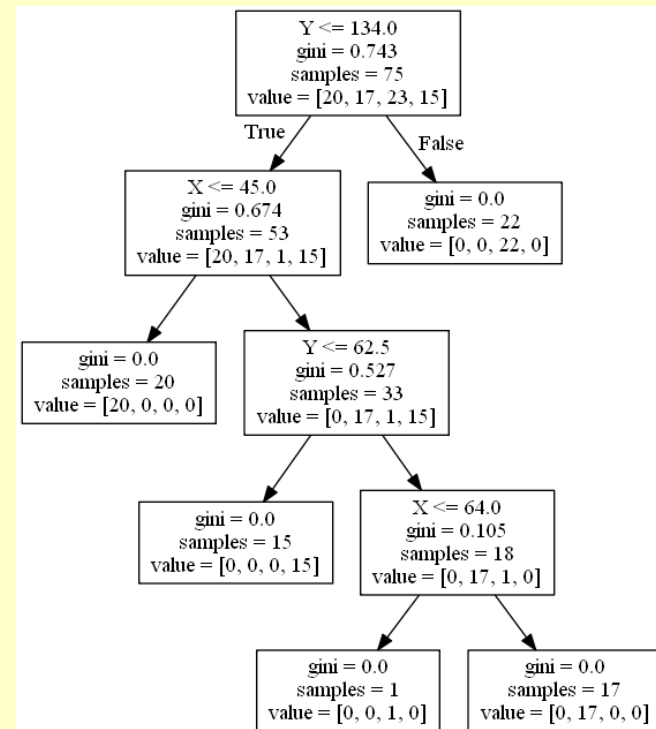
```
acc=dtc.score(train_data, train_label)
err=round((1-acc)*100, 2)
print("\n\nError ratio = ', err, '%')
```

```
dot_data = tree.export_graphviz(dtc, out_file=None,
feature_names=train_data.columns.values)
```

```
graph = graphviz.Source(dot_data, format="png")
graph.render( view=True)
```

Class = [4]

Error ratio = 0.0 %



Latihan 6 - #Assignment

(Decision Tree dengan nilai numerik)

1. `dataset ← titanic.csv`
2. `test_dataset ← titanic_test.csv`
3. `train_data ← ambil dataset kolom fitur (Sex, Age, Pclass, Fare).`
Lakukan pengisian missing value pada fitur Age dengan nilai mean dari masing-masing class
4. `test_data ← ambil test_dataset kolom fitur (Sex, Age, Pclass, Fare).`
5. `train_label ← ambil dataset kolom kelas (Survived)`
6. `test_label ← titanic_testlabel.csv`
7. Lakukan klasifikasi `test_data` terhadap `train_data` dengan Decision Tree, dan berapakah error rasionya?
8. Tampilkan hirarki dari Decision Tree

Pengumpulan Tugas

- Buatlah coding dengan Bahasa pemrograman/tools apapun untuk semua assignment
- Buatlah laporan dalam slide ppt. Laporan terdiri dari screenshot coding dan hasil running untuk setiap assignment.
- Simpan laporan dalam file pdf dengan format penamaan:
DM_M6_NRP_namadepan.pdf
- Deadline upload:
 - Reguler & RPL Kelas A : Selasa, 29 September 2026 pk 23.00
 - RPL Kelas B : Sabtu, 3 Oktober 2026 pk 23.00